Advanced Programming Practice · 21CSC203P

Unit 3 · Session 8 · SLO 1 — Solved Worksheet

Name: AKARSH MURALI AKARSH MURALI | **Reg. No:** RA2511030010067

Unit – 3 : Session – 8 : SLO – 1 | Apply Exception Handling in Database Access

I. Fill in the Blanks

1. SQLException — it is the base checked exception for all JDBC database access errors.
2. finally — the finally block (or try-with-resources) ensures database resources are closed properly.
3. AutoCloseable — try-with-resources automatically closes any object that implements the AutoCloseable interface.
4. subclass–superclass (inheritance) — in a multi-catch block the exception types must not be related by inheritance.

II. True or False

1. TRUE — SQLException provides getMessage(), getSQLState() and getErrorCode() for diagnosing database errors.
2. TRUE — ResultSet and Statement must be explicitly closed in a finally block, or automatically via try-with-resources.

III. Multiple Choice

1. B — catch(SQLException e) is the correct syntax for catching database-related exceptions.
2. C — getSQLState() returns a standardized five-character SQL error code string.

IV. Programming

1. Identify the output: A — The connection fails, the SQLException is caught, and the program prints "Error: " followed by the driver-supplied message (e.g. invalid URL / connection refused).
2. Identify the error: B — executeQuery() throws SQLException, not IOException; the catch clause must declare SQLException (or Exception), otherwise the program does not compile.